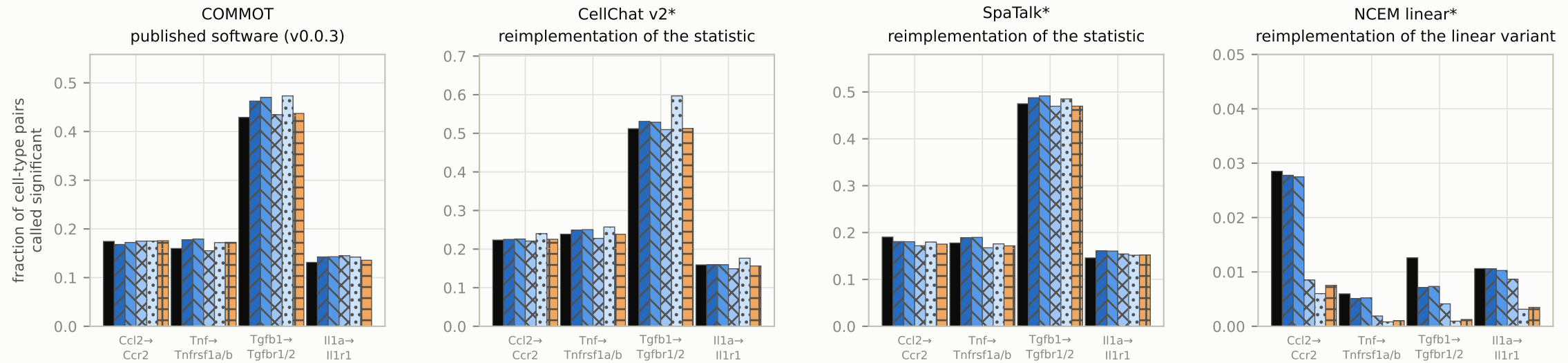


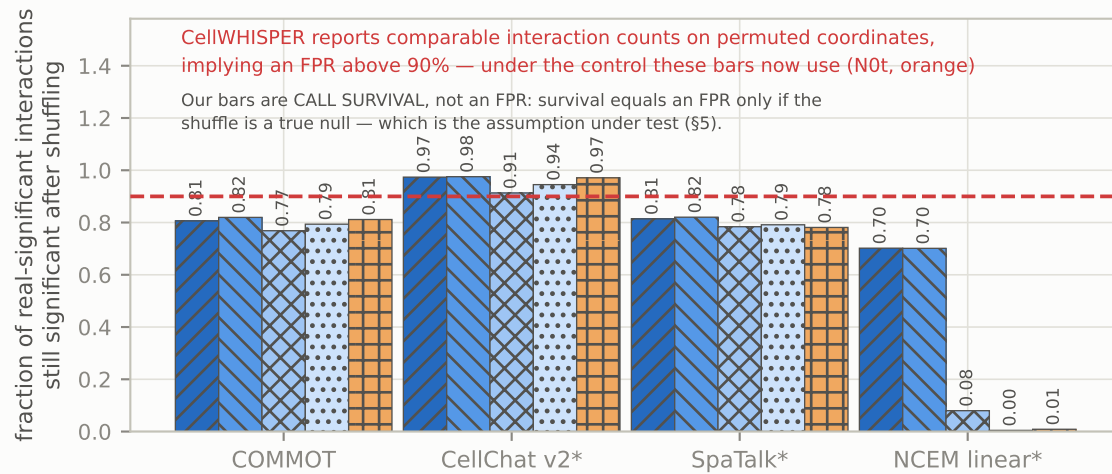
Existing spatial cell-cell-communication tools do not distinguish real tissue from coordinate-shuffled tissue

6 admissible sections × 3 tiles × 4 senescence-relevant ligand-receptor pairs; sender × receiver cell-type interactions; significance at $\alpha = 0.05$ (NCEM: BH-FDR $q < 0.05$)

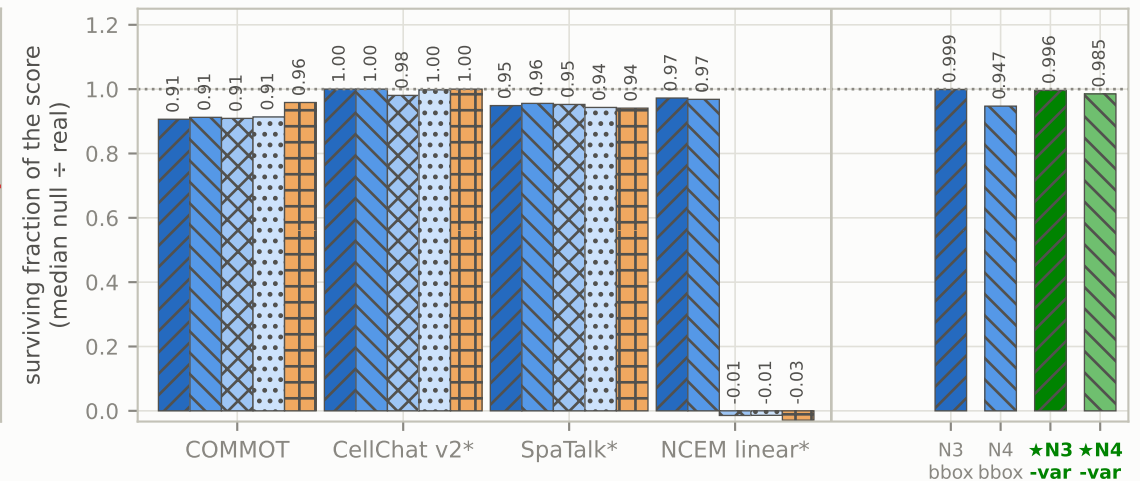
a Interactions called significant, real vs shuffled coordinates



b Does the coordinate shuffle change what is CALLED?



c Does the coordinate shuffle change the SCORE?



our SASP kernel estimator: ★ = the frozen primary (variance-corrected) null

real coordinates
 N4 rotation, ligand⁺ cells
 N3t torus shift, per cell type
 N0 full coordinate permutation
 N0t within-cell-type permutation (CellWHISPER's own control)
 N3 torus shift, ligand⁺ cells

* reimplementations of the published statistics, NOT the published software. CellChat v2 and SpaTalk are R packages and NCEM needs Python ≤ 3.10 + TensorFlow; only COMMOT ran as released software.

COMMOT is run on 1.2 mm tiles at native density because commot 0.0.3 materialises a dense $n \times n$ distance matrix (105 GB for a whole section).

CellChat is run with `type="mean"`: at its default `triMean` the ligand term is exactly zero for 3 of the 4 pairs on this panel, and it calls nothing at all.